Supplementary Report: Two-Agent Generation of Positive Examples

Source-grounded question writing, blind verification, and reproducible export for DPrivBench

S1 Scope

We generate positive (`label=1`) DPrivBench items from a fixed corpus of differential privacy papers [1–5]. Each item is a closed claim whose correct answer is Yes and includes a checkable, source-anchored proof. A two-agent protocol separates writing from verification, while deterministic checks enforce schema validity, hash consistency, LaTeX balance, unique identifiers, and exact-text deduplication.

S2 Generation and Verification Protocol

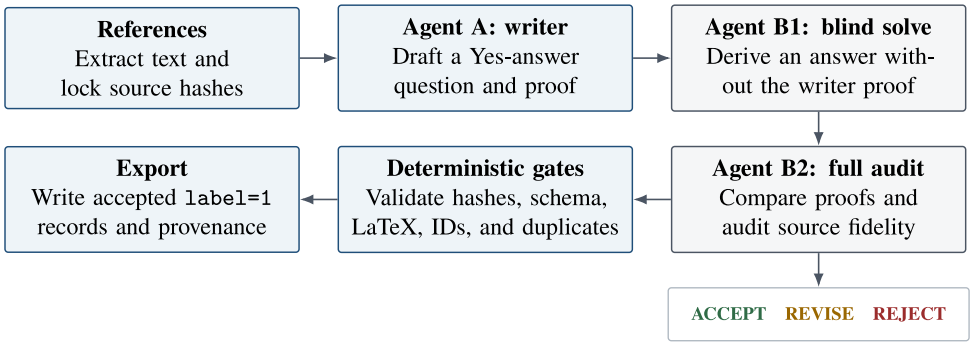


Figure S1: Positive-example workflow. B1 receives neither Agent A's proof nor its source evidence. A revised candidate returns to Agent A and undergoes a new B1 solve and B2 audit; rejected candidates are not exported.

Agent A may use only the locked manifest, extracted source text, and generator schema. It preserves the source assumptions, quantifiers, adjacency relation, privacy notion, parameter domains, mechanism, and constants. Agent B first solves the proof-hidden question (B1), then audits the complete candidate against the immutable B1 record and the locked references (B2).

S3 Models and Information Barriers

Both roles used `gpt-5.6-sol` with `high` reasoning effort (Table S1). Using the same model controls capability across stages. Verification independence is provided by separate contexts, restricted inputs, proof hiding, and hash locking, not by model heterogeneity.

Table S1: Agent roles and visible information.

Stage	Model	Visible input	Output
Agent A	`gpt-5.6-sol`, high	Locked manifest, extracted sources, generator schema	Question, public proof, assumptions, and source anchors
Agent B1	`gpt-5.6-sol`, high	Proof-hidden question view and locked sources	Independent Yes/No/Undetermined answer, proof or counterexample, and anchors
Agent B2	`gpt-5.6-sol`, high	Full candidate, immutable B1 analysis, and locked sources	Adversarial audit and ACCEPT/REVISE/REJECT verdict

S4 Acceptance Criteria

A candidate is exported only if B1 independently answers Yes, every B1 and B2 hard check passes, and neither stage reports a major or fatal issue. Agent A's proof must be mathematically valid and source-faithful; assumptions, parameters, adjacency, privacy notion, constants, quantifiers, and source locators must all match the references. The question, proof, B1 analysis, and manifest hashes must also match the locked artifacts. Any substantive change requires a new verification cycle.

S5 Experimental Configuration

Table S2 reports the recorded configuration for run `positive-20260729-001`. The corpus was processed as one locked batch, with candidates and verification records tracked by source and candidate identifier.

Table S2: Configuration of the recorded generation run.

Field	Value
Run initialized	29 July 2026, 23:46 UTC
Reference corpus	5 local PDF papers; 142 pages total
Extraction	High quality for all 5 papers
Access policy	Locked local references only; no web browsing or corpus expansion
Generation unit	Closed, source-supported Yes-answer claim with a public proof and a precise theorem, lemma, proposition, corollary, equation, page, or section anchor
Verification	Proof-hidden B1 solve; immutable B1 hash; B2 full audit; deterministic schema, LaTeX, hash, ID, and duplicate checks

S6 Results

Table S3: Source-level generation and acceptance counts.

Source	Pages	Extraction	Written	Accepted
Koskela et al. [1]	26	High	2	2
Lebeda [2]	15	High	2	2
Tran et al. [3]	45	High	2	2
Haghifam et al. [4]	40	High	2	2
Joseph et al. [5]	16	High	2	2
Total	**142**	**5/5 High**	**10**	**10**

Agent A wrote 10 candidates, two per reference. B1 independently returned Yes for 10/10 candidates, with all blind checks passing. B2 assigned ACCEPT to 10/10 candidates; all hard checks passed and no B1 or B2 issues were recorded. The exporter produced 10 `label=1` records with question identifiers 1–10. These results describe this run on the selected corpus and should not be interpreted as an estimate of performance on unseen literature.

S7 Reproducibility and Review

Source PDFs and extracted text are SHA-256 hashed; provenance links each export to its source, question, proof, B1 analysis, and B2 audit hashes. Configurations are stored in `.codex/agents/`, and run artifacts in `runs/positive-20260729-001/`.

Automated gates do not replace curation. We recommend reviewing at least 10% of each batch and every candidate with weak extraction, a multi-theorem derivation, nonstandard adjacency or privacy conversion, divergent proof routes, or a revision.

References

[1] A. Koskela, R. Redberg, and Y.-X. Wang. Privacy profiles for private selection. arXiv:2402.06701, 2024.
[2] C. J. Lebeda. Better Gaussian mechanism using correlated noise. arXiv:2408.06853, 2025.
[3] L. Tran, A. Koskela, O. Räisä, and A. Honkela. f-differential privacy filters: Validity and approximate solutions. arXiv:2602.06756, 2026.
[4] M. Haghifam, T. Steinke, and J. Ullman. Private geometric median. NeurIPS, 2024.
[5] M. Joseph, A. Kulesza, and A. Yu. Approximate differential privacy of the ℓ_2 mechanism. ICML, PMLR 267, 2025.